

FUNDAMENTALS OF MECHANICAL ENGINEERING

Mechanical engineering is a branch of engineering that deals with the design, construction, and use of machines. It applies the principles of engineering, physics, and materials science to the design, analysis, manufacturing, and maintenance of mechanical systems.

Some of the fundamental concepts in mechanical engineering include:

1. **Mechanics:** The study of forces and their effect on matter. This includes the study of statics, dynamics, and kinematics.
2. **Thermodynamics:** The study of heat and its relationship to work and energy. This includes the study of heat transfer, thermodynamic cycles, and the laws of thermodynamics.
3. **Materials Science:** The study of the properties and behavior of materials. This includes the study of the structure, properties, and performance of materials.
4. **Manufacturing:** The process of converting raw materials into finished products. This includes the study of manufacturing processes, production planning, and quality control.
5. **Design:** The process of creating a plan or drawing to show the look and function of an object before it is made. This includes the study of design principles, computer-aided design, and product design.

Unit 1 - Introduction to Mechanics

Mechanics is the branch of physics that deals with the motion of objects and the forces that cause that motion. It is divided into two main parts: kinematics and dynamics.

1. **Kinematics** is the study of motion without considering the forces that cause it. It describes how objects move in terms of position, velocity, and acceleration.
2. **Dynamics** is the study of motion and the forces that cause it. It explains why objects move the way they do and how to predict their future motion.

Some important concepts in mechanics include: - **Displacement:** the change in position of an object. - **Velocity:** the rate of change of displacement with respect to time. - **Acceleration:** the rate of change of velocity with respect to time. - **Force:** a push or pull on an object that can cause it to accelerate. - **Mass:** a measure of the amount of matter in an object. - **Weight:** the force of gravity on an object. - **Newton's laws of motion:** three laws that describe the relationship between the motion of an object and the forces acting on it.

Mechanics has many practical applications, including the design of machines, vehicles, and structures, as well as the study of the motion of planets and other celestial bodies. It is a fundamental subject in physics and engineering.

Force moment and couple

Force moment

- The moment of a force is a measure of its tendency to cause a body to rotate about a specific point or axis.
- It is calculated as the product of the force and the perpendicular distance from the line of action of the force to the point or axis of rotation.
- The unit of moment of force is newton-meter (N.m) in the SI system.

Couple

- A couple is a pair of forces, equal in magnitude, oppositely directed, and displaced by perpendicular distance or moment.
- The effect of a couple is to produce rotation without translation.
- The moment of a couple is the same about any point and is equal to the product of one of the forces and the perpendicular distance between the forces.
- The unit of moment of a couple is also newton-meter (N.m) in the SI system.

Principle of Transmissibility

The principle of transmissibility states that the external effect of a force on a rigid body is the same, regardless of the point of application, provided the line of action of the force remains unchanged. This means that the force can be transmitted through the body without changing the state of rest or motion of the body.

This principle is important in the study of mechanics as it allows us to analyze the forces acting on a body without having to consider the exact point of application of the force. Instead, we can focus on the line of action of the force and its magnitude.

Here are some key points to remember about the principle of transmissibility: - The external effect of a force on a rigid body is the same, regardless of the point of application, provided the line of action of the force remains unchanged. - The force can be transmitted through the body without changing the state of rest or motion of the body. - This principle allows us to analyze the forces acting on a body without having to consider the exact point of application of the force. - Instead, we can focus on the line of action of the force and its magnitude.

This principle is an important concept in the study of mechanics and is covered in Unit 1 - Introduction to Mechanics in the subject of FUNDAMENTALS OF

MECHANICAL ENGINEERING. It is essential to have a good understanding of this principle to be able to analyze and solve problems in mechanics.

Varignon's Theorem

Varignon's Theorem is a statement in mechanics that deals with the moment of a force. It states that the moment of a force about a point is equal to the algebraic sum of the moments of the components of the force about the same point.

In other words, if a force is resolved into two or more components, the moment of the force about a point is equal to the sum of the moments of the components about the same point.

This theorem is useful in solving problems involving the moment of a force, as it allows us to break down a complex force into simpler components and calculate their moments separately.

Here are the key points to remember about Varignon's Theorem: - It applies to the moment of a force about a point. - The moment of a force is equal to the sum of the moments of its components about the same point. - This theorem is useful in solving problems involving the moment of a force.

Resultant of force system- concurrent and non-concurrent coplanar forces

Unit 1 - Introduction to Mechanics

FUNDAMENTALS OF MECHANICAL ENGINEERING

1. A force system is a collection of forces acting on a body or a group of bodies.
2. The resultant of a force system is a single force that has the same effect on the body as the original force system.
3. Concurrent forces are forces that act on a body at the same point.
4. Non-concurrent forces are forces that act on a body at different points.
5. Coplanar forces are forces that lie in the same plane.
6. The resultant of a concurrent coplanar force system can be found by adding the forces vectorially.
7. The resultant of a non-concurrent coplanar force system can be found by finding the moment of each force about a point and then finding the resultant moment. The line of action of the resultant force passes through the point where the resultant moment is zero.
8. The magnitude and direction of the resultant force can be found using the principles of statics.

Unit 1 - Introduction to Mechanics

Types of Supports and Loads

In the study of mechanics, it is important to understand the different types of supports and loads that can be applied to a structure.

Supports

- **Hinge support:** A hinge support allows rotation but prevents translation in any direction. It is also known as a pin support.
- **Roller support:** A roller support allows rotation and translation in one direction, but prevents translation in the other direction.

Loads

- **Point load:** A point load is a load that acts at a single point on a structure.
- **Uniformly Distributed Load (UDL):** A UDL is a load that is distributed evenly over a length of a structure.
- **Uniformly Varying Load (UVL):** A UVL is a load that varies linearly over a length of a structure.

These are some of the basic types of supports and loads that are commonly used in the study of mechanics. Understanding these concepts is essential for analyzing and designing structures.

Free Body Diagram

A free body diagram is a graphical representation of the forces acting on an object. It is used to analyze the forces acting on an object and to determine the net force and the resulting acceleration.

Here are some key points to remember when constructing a free body diagram:

1. Draw a diagram of the object, representing it as a point mass or a simple shape.
2. Identify all the forces acting on the object and represent them as vectors, with the tail of the vector at the point of application of the force and the head of the vector pointing in the direction of the force.
3. Label each force with its type (e.g. gravitational force, normal force, frictional force, etc.) and its magnitude (if known).
4. The net force acting on the object is the vector sum of all the individual forces. This can be calculated by adding the vectors head-to-tail or by using the Pythagorean theorem and trigonometry.
5. The acceleration of the object is determined by the net force and the mass of the object, according to Newton's second law of motion: $F_{\text{net}} = ma$.

Free body diagrams are a useful tool in solving problems in mechanics and can help to visualize and understand the forces acting on an object. It is important to carefully identify all the forces and to accurately represent their magnitudes and directions in the diagram.

Equilibrium Equations and Support Reactions

In the subject of Fundamentals of Mechanical Engineering, Unit 1 - Introduction to Mechanics, equilibrium equations and support reactions are important concepts to understand.

1. Equilibrium equations are mathematical expressions that describe the balance of forces and moments acting on a body or a system of bodies.
2. These equations are based on Newton's laws of motion, which state that an object will remain at rest or in uniform motion in a straight line unless acted upon by an external force.
3. In two-dimensional problems, there are three equilibrium equations: the sum of the forces in the x-direction must be zero, the sum of the forces in the y-direction must be zero, and the sum of the moments about any point must be zero.
4. In three-dimensional problems, there are six equilibrium equations: the sum of the forces in the x, y, and z directions must be zero, and the sum of the moments about the x, y, and z axes must be zero.
5. Support reactions are the forces and moments that a support exerts on a body to keep it in equilibrium.
6. Common types of supports include fixed supports, pinned supports, and roller supports.
7. The type of support determines the number and direction of the support reactions.
8. To solve for the support reactions, the equilibrium equations are used in conjunction with the free body diagram of the body or system of bodies.

Normal and Shear Stress

Introduction to Mechanics - Unit 1: Fundamentals of Mechanical Engineering

- **Stress** is defined as the internal resistance offered by a body to an external force.
- It is a measure of the intensity of the internal forces acting within a deformable body.
- Stress is expressed in terms of force per unit area.
- There are two main types of stress: **normal stress** and **shear stress**.

Normal Stress

- Normal stress is the stress that acts perpendicular to the plane of the cross-section of the body.
- It is denoted by the symbol σ (sigma).
- Normal stress can be either tensile or compressive.
- Tensile stress is the stress that tends to pull the particles of the body apart.

- Compressive stress is the stress that tends to push the particles of the body together.

Shear Stress

- Shear stress is the stress that acts parallel to the plane of the cross-section of the body.
- It is denoted by the symbol τ (tau).
- Shear stress is the stress that tends to cause the particles of the body to slide over one another.
- It is the stress that causes deformation by shear strain.

Conclusion

- Normal and shear stress are two main types of stress that act on a deformable body.
- Normal stress acts perpendicular to the plane of the cross-section of the body, while shear stress acts parallel to it.
- Normal stress can be either tensile or compressive, while shear stress causes deformation by shear strain.

Strain

Strain is a measure of deformation representing the displacement between particles in the body relative to a reference length. It is defined as the change in length divided by the original length. Strain is a dimensionless quantity and is usually expressed as a decimal fraction, a percentage, or in parts-per notation.

There are two types of strain: normal strain and shear strain.

- Normal strain: Normal strain occurs when the deformation of a body is caused by an external force that is perpendicular to the surface of the body. It is defined as the change in length divided by the original length.
- Shear strain: Shear strain occurs when the deformation of a body is caused by an external force that is parallel to the surface of the body. It is defined as the change in the angle between two lines that were originally perpendicular.

Strain can be either positive or negative. Positive strain indicates that the body is elongated, while negative strain indicates that the body is compressed.

Strain is an important concept in mechanics, as it is used to determine the amount of deformation a body undergoes when subjected to external forces. It is also used to calculate the stress in a body, which is the force per unit area that arises from the deformation of the body.

Hooke's law

Hooke's law is a principle of physics that states that the force needed to extend or compress a spring by some distance is proportional to that distance. It is expressed mathematically as $F = kx$, where F is the force applied to the spring, x is the displacement of the spring from its equilibrium position, and k is the spring constant, a measure of the stiffness of the spring.

- Hooke's law applies to many materials, not just springs, and is a fundamental principle of elasticity.
- The spring constant, k , is a measure of the stiffness of the spring and is determined by the material and geometry of the spring.
- The displacement, x , is the change in length of the spring from its equilibrium position.
- Hooke's law is an idealization and is only approximately true for small displacements. For larger displacements, the force-displacement relationship may become nonlinear.
- Hooke's law is named after the 17th-century British physicist Robert Hooke, who first formulated the law in 1660.

Poisson's ratio

Poisson's ratio is a measure of the deformation of a material when subjected to an applied stress. It is defined as the ratio of the lateral strain to the axial strain in the direction of the applied stress.

- Poisson's ratio is denoted by the Greek letter nu (ν).
- It is a dimensionless quantity, meaning it has no units.
- The value of Poisson's ratio typically ranges between -1 and 0.5 for most materials.
- A value of 0.5 indicates that the material is incompressible, meaning that when it is subjected to an applied stress, its volume remains constant.
- A value of 0 indicates that the material is perfectly compressible, meaning that when it is subjected to an applied stress, its volume changes but its shape remains unchanged.
- A negative value of Poisson's ratio indicates that the material exhibits auxetic behavior, meaning that when it is subjected to an applied stress, it expands in the direction perpendicular to the applied stress.

Poisson's ratio is an important property of materials as it is used in the calculation of stress and strain in various engineering applications. It is also used in the design of structures and components to ensure that they can withstand the applied loads without failure.

Elastic Constants and their Relationship

Elastic constants are the material properties that describe the relationship between stress and strain in a material. There are several elastic constants, in-

cluding:

1. **Young's Modulus (E):** This is the ratio of tensile stress to tensile strain within the elastic limit. It is a measure of the stiffness of a material.
2. **Shear Modulus (G):** This is the ratio of shear stress to shear strain within the elastic limit. It is a measure of the rigidity of a material.
3. **Bulk Modulus (K):** This is the ratio of hydrostatic stress to volumetric strain within the elastic limit. It is a measure of the compressibility of a material.
4. **Poisson's Ratio (ν):** This is the ratio of lateral strain to axial strain within the elastic limit. It is a measure of the degree to which a material tends to expand in directions perpendicular to the direction of compression.

These elastic constants are related to each other through various equations. For example, the relationship between Young's Modulus, Shear Modulus, and Poisson's Ratio is given by the equation:

$$E = 2G(1 + \nu)$$

The relationship between Bulk Modulus, Shear Modulus, and Poisson's Ratio is given by the equation:

$$K = 2G(1 + \nu) / (3(1 - 2\nu))$$

These relationships can be used to determine the values of one elastic constant if the values of the others are known. They are important in the study of mechanics and the behavior of materials under stress.

Stress-Strain Diagram for Ductile and Brittle Materials

A stress-strain diagram is a graphical representation of the relationship between the stress and strain of a material. It is used to determine the mechanical properties of a material, such as its strength, ductility, and toughness.

Ductile Materials

- Ductile materials are materials that can undergo large plastic deformation before failure.
- The stress-strain diagram for a ductile material typically has a linear elastic region, followed by a non-linear plastic region.
- In the elastic region, the material obeys Hooke's Law, where the stress is proportional to the strain.
- In the plastic region, the material undergoes permanent deformation.
- The point at which the material transitions from elastic to plastic behavior is called the yield point.
- The maximum stress that the material can withstand before failure is called the ultimate tensile strength.

Brittle Materials

- Brittle materials are materials that undergo little to no plastic deformation before failure.
- The stress-strain diagram for a brittle material typically has a linear elastic region, followed by a sudden fracture.
- In the elastic region, the material obeys Hooke's Law, where the stress is proportional to the strain.
- The maximum stress that the material can withstand before failure is called the ultimate tensile strength.
- Unlike ductile materials, brittle materials do not have a distinct yield point.

Factor of Safety

Factor of safety (FoS) is an engineering term that describes how much stronger a system or structure is than it is required to be to fulfill its purpose under expected conditions . There are two definitions for the factor of safety:

1. The ratio of a structure's absolute strength (structural capability) to actual applied load; this is a measure of the reliability of a particular design. This is a calculated value, and is sometimes referred to, for the sake of clarity, as a realized factor of safety .
2. A constant required value, imposed by law, standard, specification, contract or custom, to which a structure must conform .

A simple calculation for safety factor is; $\text{Factor of Safety} = \frac{\text{Actual Breaking Strength (lbs)}}{\text{Normal Working Load (lbs)}}$. The term "actual breaking strength" refers to the amount of load that will cause the system to fail .

Factor of safety exists as a safety measure that is designed to make a product, a system, or a structure more reliable and less likely to fail .

Unit 2 - Introduction to IC Engines and Electric Vehicles

IC Engines

1. An internal combustion engine (IC engine) is a type of heat engine where the combustion of fuel occurs within the engine, producing mechanical work.
2. The most common types of IC engines are the reciprocating engine and the rotary engine.
3. In a reciprocating engine, the combustion of fuel takes place in a cylinder, and the expanding gases push a piston, which in turn rotates a crankshaft to produce mechanical work.
4. In a rotary engine, the combustion of fuel takes place in a chamber, and the expanding gases push against a rotor, which rotates to produce mechanical work.

5. IC engines can be powered by a variety of fuels, including gasoline, diesel, and natural gas.

Electric Vehicles

1. An electric vehicle (EV) is a vehicle that is powered by an electric motor, rather than an internal combustion engine.
2. EVs can be powered by a variety of sources, including batteries, fuel cells, and solar panels.
3. The most common type of EV is the battery electric vehicle (BEV), which is powered by a rechargeable battery.
4. EVs produce zero tailpipe emissions, making them a more environmentally friendly option than vehicles powered by internal combustion engines.
5. The range of an EV depends on the capacity of its battery, and can vary widely between different models.

IC Engine

An internal combustion engine (IC engine) is a type of heat engine where the combustion of fuel occurs within the engine, producing mechanical work. This mechanical work is used to power vehicles, machines, and other devices. The most common types of IC engines are the spark ignition (SI) engine, which is used in gasoline-powered vehicles, and the compression ignition (CI) engine, which is used in diesel-powered vehicles.

Some key points to remember about IC engines are:

1. The combustion of fuel occurs within the engine, producing mechanical work.
2. The most common types of IC engines are the spark ignition (SI) engine and the compression ignition (CI) engine.
3. SI engines are used in gasoline-powered vehicles, while CI engines are used in diesel-powered vehicles.
4. IC engines are used to power vehicles, machines, and other devices.

Basic definition of engine and Components for the notes of the Unit 2 - Introduction to IC Engines and Electric Vehicles in the subject of FUNDAMENTALS OF MECHANICAL ENGINEERING

An engine is a machine designed to convert one form of energy into mechanical energy. Internal combustion engines (IC engines) are a type of engine that burn fuel within the engine itself to produce mechanical energy. Electric vehicles, on the other hand, use electric motors to convert electrical energy into mechanical energy.

The main components of an IC engine include: - Cylinder: The cylinder is the main body of the engine where the fuel is burned and the piston moves up and down. - Piston: The piston is a cylindrical component that moves up and down

within the cylinder, compressing the fuel-air mixture and transmitting the force of the explosion to the crankshaft. - Crankshaft: The crankshaft converts the linear motion of the piston into rotational motion, which is used to power the wheels of the vehicle. - Fuel system: The fuel system delivers fuel to the engine, where it is mixed with air and burned to produce energy. - Cooling system: The cooling system helps to regulate the temperature of the engine, preventing it from overheating. - Exhaust system: The exhaust system removes the waste gases produced by the combustion process from the engine.

Unit 2 - Introduction to IC Engines and Electric Vehicles

Construction and Working of Two stroke and four stroke SI & CI engine

- An internal combustion engine (ICE) is a heat engine where the combustion of a fuel occurs with an oxidizer (usually air) in a combustion chamber that is an integral part of the working fluid flow circuit.
- There are two main types of internal combustion engines: spark ignition (SI) and compression ignition (CI).
- SI engines use a spark plug to ignite the fuel-air mixture, while CI engines rely on the heat generated by the compression of the air in the cylinder to ignite the fuel.
- Both SI and CI engines can be designed to operate on either a two-stroke or a four-stroke cycle.
- In a two-stroke engine, the four events of intake, compression, power, and exhaust occur in two strokes of the piston (one up and one down) for each cycle of the engine.
- In a four-stroke engine, these four events occur in four strokes of the piston (two up and two down) for each cycle of the engine.
- The main components of an internal combustion engine include the cylinder, piston, connecting rod, crankshaft, and valves.
- The construction and operation of a two-stroke and four-stroke engine differ in several ways.
- In a two-stroke engine, the intake and exhaust events occur simultaneously as the piston moves towards the bottom of the cylinder, while in a four-stroke engine, these events occur separately during the intake and exhaust strokes.
- Additionally, two-stroke engines require a method of lubrication, such as oil mixed with the fuel, while four-stroke engines have a separate lubrication system.
- The choice between a two-stroke and four-stroke engine depends on the specific application and the desired balance between power, efficiency, and emissions.

Unit 2 - Introduction to IC Engines and Electric Vehicles

Merits of IC Engines:

1. IC engines have a high power-to-weight ratio, making them suitable for use in vehicles.
2. They are highly reliable and have a long lifespan.
3. IC engines can be easily refueled, making them convenient for long-distance travel.
4. They are widely available and have a well-established infrastructure for fueling and maintenance.

Demerits of IC Engines:

1. IC engines produce harmful emissions, contributing to air pollution and climate change.
2. They are less efficient than electric vehicles, resulting in higher fuel costs.
3. IC engines require regular maintenance, which can be costly.
4. They are noisy and can contribute to noise pollution.

Merits of Electric Vehicles:

1. Electric vehicles produce zero tailpipe emissions, making them environmentally friendly.
2. They are highly efficient, resulting in lower operating costs.
3. Electric vehicles require less maintenance than IC engines, reducing long-term costs.
4. They are quiet and smooth to drive.

Demerits of Electric Vehicles:

1. Electric vehicles have a limited range and require regular charging, which can be inconvenient.
2. The charging infrastructure is not as well-established as the fueling infrastructure for IC engines.
3. Electric vehicles have a higher upfront cost than IC engines.
4. The production of batteries for electric vehicles has environmental impacts.

Scavenging Process

Scavenging is the process that occurs in an internal combustion engine during the overlapping of valves (when they open at the same time) in which burnt gases are released out of the cylinder with the entering of fresh charge inside the engine at the same time. The good scavenging process is significant for the engine's proper fuel combustion and better power output.

There are three types of scavenging used in two-stroke IC engines : 1. Uniflow scavenging 2. Cross-flow scavenging 3. Loop or reverse scavenging

If scavenging is incomplete, the remaining exhaust gases can cause improper combustion for the next cycle, leading to reduced power output.

Unit 2 - Introduction to IC Engines and Electric Vehicles

FUNDAMENTALS OF MECHANICAL ENGINEERING

Difference between Two-Stroke and Four-Stroke IC Engines:

1. **Cycle completion:** A two-stroke engine completes a power cycle in two strokes of the piston (one up and one down movement) while a four-stroke engine completes a power cycle in four strokes of the piston (two up and two down movements).
2. **Valve mechanism:** Two-stroke engines do not have a dedicated valve mechanism while four-stroke engines have a complex valve mechanism to control the intake and exhaust of gases.
3. **Power output:** Two-stroke engines have a higher power-to-weight ratio and can produce more power with a smaller engine size compared to four-stroke engines.
4. **Fuel efficiency:** Four-stroke engines are generally more fuel-efficient than two-stroke engines.
5. **Lubrication:** Two-stroke engines require a mixture of oil and fuel for lubrication while four-stroke engines have a separate lubrication system.
6. **Emissions:** Two-stroke engines generally produce more emissions than four-stroke engines due to incomplete combustion of the fuel-oil mixture.

Difference between SI and CI Engines:

1. **Ignition:** SI engines use a spark plug to ignite the air-fuel mixture while CI engines use the heat generated by the compression of the air-fuel mixture to ignite it.
2. **Fuel:** SI engines generally use gasoline as fuel while CI engines use diesel.
3. **Compression ratio:** CI engines have a higher compression ratio than SI engines.
4. **Efficiency:** CI engines are generally more efficient than SI engines due to their higher compression ratio.
5. **Noise:** CI engines are generally noisier than SI engines due to the higher pressure and temperature of the combustion process.
6. **Emissions:** CI engines generally produce less CO and HC emissions than SI engines but produce more NOx and particulate matter emissions.

Electric vehicles and hybrid vehicles

Electric vehicles (EVs) and hybrid vehicles are two types of vehicles that are powered by electricity. They differ in their fuel type and the way they generate electricity.

- **Electric vehicles (EVs)** have a battery instead of a gasoline tank, and an electric motor instead of an internal combustion engine. The battery can be recharged with electricity, which comes from many generating sources,

from the burning of fossil fuels to renewable sources like solar, wind, and hydro power.

- **Hybrid vehicles** are a combination of gasoline and electric vehicles, so they have a battery, an electric motor, a gasoline tank, and an internal combustion engine. They are entirely fueled by gasoline and derive the electricity they need through brake regeneration and the harnessing of excess engine power.
- **Plug-in hybrid electric vehicles (PHEVs)** are a type of hybrid vehicle that can be recharged by plugging into an external source of electricity, as well as by their internal combustion engine.

These vehicles offer a more environmentally friendly and sustainable alternative to traditional gasoline-powered vehicles. They produce fewer emissions and can be more cost-effective to operate in the long run. However, the availability of charging infrastructure and the initial cost of the vehicles can be barriers to their widespread adoption.

Components of an EV

An electric vehicle (EV) is a vehicle that is powered by an electric motor and a rechargeable battery. The main components of an EV are:

1. **Electric Motor:** The electric motor is the primary source of propulsion for an EV. It converts electrical energy into mechanical energy to drive the wheels of the vehicle.
2. **Battery:** The battery is the primary source of electrical energy for an EV. It stores electrical energy and provides it to the electric motor to power the vehicle.
3. **Power Electronics:** The power electronics in an EV control the flow of electrical energy between the battery and the electric motor. They also manage the charging of the battery.
4. **Transmission:** The transmission in an EV transfers the mechanical energy from the electric motor to the wheels of the vehicle. It can be a single-speed or multi-speed transmission.
5. **Charging System:** The charging system in an EV allows the battery to be recharged from an external power source. It can be a slow or fast charging system.
6. **Regenerative Braking:** Regenerative braking is a system that captures the kinetic energy of the vehicle during braking and converts it into electrical energy to recharge the battery.
7. **Thermal Management System:** The thermal management system in an EV manages the temperature of the battery, electric motor, and power electronics to ensure optimal performance and longevity.

These are the main components of an EV. They work together to provide a smooth, efficient, and environmentally friendly driving experience.

EV Batteries

- EV batteries are used to power electric vehicles.
- The most common battery type in modern electric vehicles are lithium-ion and lithium polymer.
- Other types of rechargeable batteries used in electric vehicles include lead-acid, nickel-cadmium, nickel-metal hydride, zinc-air, and sodium nickel chloride.
- Most electric vehicles are powered by lithium-ion batteries and regenerative braking, which slows a vehicle down and generates electricity at the same time.
- The types of EVs that use batteries include all-electric vehicles, also known as battery electric vehicles (BEVs), which are completely powered by electricity.
- Electric-car batteries typically weigh around 1,000 pounds, cost around \$15,000 to manufacture, and have enough power to run a typical home for a few days.

Unit 2 - Introduction to IC Engines and Electric Vehicles

Chargers

- Chargers are devices used to replenish the energy stored in rechargeable batteries.
- Chargers for electric vehicles (EVs) come in different types, with varying power levels and charging speeds.
- Level 1 chargers, also known as trickle chargers, use a standard 120V AC outlet and can take several hours to fully charge an EV battery.
- Level 2 chargers use a 240V AC outlet and can charge an EV battery in a few hours.
- DC fast chargers, also known as Level 3 chargers, use direct current (DC) and can charge an EV battery to 80% capacity in less than an hour.
- The type of charger used can affect the charging time and cost of charging an EV.
- Some EVs come with an onboard charger, while others require an external charger to be installed.
- Charging infrastructure is an important consideration for the widespread adoption of EVs.

Drives

Drives are the means by which power is transmitted from the engine or motor to the wheels of a vehicle. There are several types of drives used in vehicles, including:

1. **Chain drive:** A chain drive is a system that uses a chain to transmit power from the engine to the wheels. It is commonly used in motorcycles and bicycles.
2. **Belt drive:** A belt drive is a system that uses a belt to transmit power from the engine to the wheels. It is commonly used in scooters and some small cars.
3. **Shaft drive:** A shaft drive is a system that uses a shaft to transmit power from the engine to the wheels. It is commonly used in cars and trucks.
4. **Electric drive:** An electric drive is a system that uses an electric motor to transmit power to the wheels. It is commonly used in electric vehicles.

Each type of drive has its own advantages and disadvantages, and the choice of drive depends on the specific requirements of the vehicle. For example, chain drives are simple and efficient, but they require regular maintenance. Belt drives are quiet and smooth, but they can slip in wet conditions. Shaft drives are durable and require little maintenance, but they are heavier and more expensive than other types of drives. Electric drives are efficient and produce zero emissions, but they require a large battery pack and charging infrastructure.

In Unit 2 of the subject Fundamentals of Mechanical Engineering, we will learn more about the different types of drives and their applications in internal combustion engines and electric vehicles.

Transmission and Power Devices

Transmission and power devices are essential components in the operation of internal combustion engines and electric vehicles. These devices are responsible for transmitting power from the engine or motor to the wheels, allowing the vehicle to move.

1. **Transmission:** The transmission is a device that transfers power from the engine to the drive wheels. It does this by using gears to change the speed and torque of the engine output. There are two main types of transmissions: manual and automatic.
 - **Manual transmission:** A manual transmission requires the driver to manually shift gears using a clutch and gear shift. The driver must disengage the clutch, select the appropriate gear, and then re-engage the clutch to transmit power to the wheels.
 - **Automatic transmission:** An automatic transmission uses a torque converter and planetary gear sets to automatically change gears based on the speed of the vehicle and the load on the engine. The driver does not need to manually shift gears.
2. **Differential:** The differential is a device that allows the drive wheels to rotate at different speeds. This is necessary when a vehicle turns, as the

wheels on the outside of the turn must rotate faster than the wheels on the inside. The differential uses gears to transfer power from the driveshaft to the wheels while allowing them to rotate at different speeds.

3. **Drive shaft:** The drive shaft is a rotating shaft that transmits power from the transmission to the differential. It is typically made of steel and is designed to withstand the high torque and rotational speeds required to transfer power from the engine to the wheels.
4. **Axle:** The axle is a shaft that connects the wheels to the differential. It is responsible for transmitting power from the differential to the wheels. There are two main types of axles: live and dead.
 - **Live axle:** A live axle is a solid shaft that connects the wheels on one side of the vehicle. It transmits power from the differential to the wheels and also supports the weight of the vehicle.
 - **Dead axle:** A dead axle does not transmit power to the wheels. It is used only to support the weight of the vehicle.

These are some of the main transmission and power devices used in internal combustion engines and electric vehicles. Understanding these devices is essential for understanding the operation of these vehicles.

Advantages and disadvantages of EVs

Electric vehicles (EVs) are becoming increasingly popular as an alternative to traditional internal combustion engine (ICE) vehicles. Here are some advantages and disadvantages of EVs:

Advantages:

- **Zero emissions:** EVs produce no tailpipe emissions, which can help reduce air pollution and improve air quality.
- **Lower operating costs:** EVs are generally cheaper to operate than ICE vehicles, as electricity is cheaper than gasoline or diesel fuel.
- **Reduced noise pollution:** EVs are much quieter than ICE vehicles, which can help reduce noise pollution in urban areas.
- **Instant torque:** EVs have instant torque, which means they can accelerate quickly and smoothly.
- **Lower maintenance costs:** EVs have fewer moving parts than ICE vehicles, which can result in lower maintenance costs.

Disadvantages:

- **Limited range:** EVs have a limited range compared to ICE vehicles, which can be a concern for some drivers.
- **Long charging times:** It can take several hours to fully charge an EV, which can be inconvenient for some drivers.

- **Limited charging infrastructure:** The charging infrastructure for EVs is still developing, which can make it difficult to find a charging station in some areas.
- **Higher upfront costs:** EVs are generally more expensive to purchase than ICE vehicles, although this cost difference is decreasing over time.
- **Battery degradation:** The battery in an EV can degrade over time, which can reduce the range of the vehicle.

These are some of the advantages and disadvantages of EVs. It is important to consider these factors when deciding whether an EV is the right choice for you.

Hybrid Electric Vehicles

- A hybrid electric vehicle (HEV) is a type of hybrid vehicle that combines a conventional internal combustion engine (ICE) system with an electric propulsion system (hybrid vehicle drivetrain) .
- The presence of the electric powertrain is intended to achieve either better fuel economy than a conventional vehicle or better performance .
- Plug-in hybrid electric vehicles (PHEVs) are a combination of gasoline and electric vehicles, so they have a battery, an electric motor, a gasoline tank, and an internal combustion engine .
- PHEVs use both gasoline and electricity as fuel sources .
- Hybrid electric vehicles are powered by an internal combustion engine and one or more electric motors, which uses energy stored in batteries .
- A hybrid electric vehicle cannot be plugged in to charge the battery. Instead, the battery is charged through regenerative braking and by the internal combustion engine .

HEV Drive Train Components

A hybrid electric vehicle (HEV) combines a conventional internal combustion engine (ICE) propulsion system with an electric propulsion system. The presence of the electric powertrain is intended to achieve either better fuel economy than a conventional vehicle or better performance. The main components of the HEV drive train are:

1. **Internal Combustion Engine (ICE):** The ICE in an HEV is typically smaller and more efficient than in a conventional vehicle. It is used to provide propulsion power and to charge the battery.
2. **Electric Motor:** The electric motor in an HEV is used to provide additional propulsion power and to recover energy during braking.
3. **Battery:** The battery in an HEV is used to store electrical energy that is generated by the ICE and recovered during braking. It is also used to provide power to the electric motor.
4. **Power Electronics:** The power electronics in an HEV are used to control the flow of electrical energy between the battery, electric motor, and ICE.

5. **Transmission:** The transmission in an HEV is used to transfer power from the ICE and electric motor to the wheels. It may be a conventional transmission or a continuously variable transmission (CVT).
6. **Regenerative Braking System:** The regenerative braking system in an HEV is used to recover energy during braking and store it in the battery.

These are the main components of the HEV drive train. Each component plays a crucial role in the overall operation of the vehicle. It is important to have a good understanding of these components in order to fully understand the operation of an HEV.

Advantages of HV (Hybrid Vehicles)

1. **Fuel Efficiency:** Hybrid vehicles combine a traditional internal combustion engine with an electric motor, which can result in increased fuel efficiency compared to traditional gasoline-powered vehicles.
2. **Reduced Emissions:** Hybrid vehicles produce fewer emissions compared to traditional gasoline-powered vehicles, which can help reduce air pollution and improve air quality.
3. **Regenerative Braking:** Hybrid vehicles use regenerative braking, which captures the energy lost during braking and uses it to recharge the battery. This can help improve fuel efficiency and reduce wear on the brakes.
4. **Performance:** Hybrid vehicles can provide improved performance compared to traditional gasoline-powered vehicles, including faster acceleration and smoother, quieter operation.
5. **Cost Savings:** Due to their increased fuel efficiency, hybrid vehicles can result in cost savings for the owner over time, as they require less fuel to operate.
6. **Incentives:** Many governments offer incentives for the purchase of hybrid vehicles, such as tax credits or rebates, which can help offset the initial cost of the vehicle.

Unit 3 - Introduction to Refrigeration and Air-Conditioning

Refrigeration and air conditioning are two closely related technologies that involve the transfer of heat from one location to another. The main difference between the two is that refrigeration is primarily used to cool and preserve food and other perishable items, while air conditioning is used to cool and dehumidify the air in a building or vehicle.

1. **Refrigeration** is the process of removing heat from a space or substance in order to lower its temperature. This is typically done by using a refrigerant, which is a substance that absorbs heat when it evaporates and releases heat when it condenses.
2. **Air conditioning** is the process of treating air in order to control its temperature, humidity, and cleanliness. This is typically done by using

a refrigerant to cool the air, and then passing the air over a cold coil to remove moisture.

3. Both refrigeration and air conditioning systems work by using the principles of thermodynamics, which state that heat always flows from a hotter object to a cooler object.
4. The main components of a refrigeration or air conditioning system are the compressor, condenser, evaporator, and expansion valve. The compressor pumps the refrigerant through the system, the condenser releases heat from the refrigerant, the evaporator absorbs heat from the air, and the expansion valve controls the flow of refrigerant.
5. Refrigeration and air conditioning systems can be powered by electricity, natural gas, or other sources of energy. They can also be designed to use renewable energy sources such as solar power.
6. There are many different types of refrigeration and air conditioning systems, including window units, central air systems, and portable units. The type of system that is best for a particular application depends on factors such as the size of the space to be cooled, the climate, and the user's budget.
7. Proper maintenance of refrigeration and air conditioning systems is important to ensure their efficient operation and to prevent breakdowns. This includes regular cleaning of the coils and filters, checking for leaks, and replacing worn or damaged parts.
8. Refrigeration and air conditioning have had a significant impact on modern society, allowing us to preserve food, keep buildings comfortable, and transport perishable goods over long distances. However, these technologies also have an environmental impact, and efforts are being made to develop more sustainable and energy-efficient systems.

Refrigeration

Refrigeration is the process of removing heat from a space or substance to lower its temperature and maintain it below the ambient temperature. It is an essential technology in modern society, used for a variety of applications such as food preservation, air conditioning, and industrial processes.

Here are some key points to note about refrigeration:

1. Refrigeration works on the principle of the refrigeration cycle, which involves the transfer of heat from a low-temperature region to a high-temperature region using a refrigerant.
2. The refrigeration cycle consists of four main components: the compressor, the condenser, the expansion valve, and the evaporator.
3. The refrigerant is compressed in the compressor, which raises its temperature and pressure. It then flows to the condenser, where it releases heat

- to the surroundings and condenses into a liquid.
4. The liquid refrigerant then flows through the expansion valve, where its pressure is reduced, causing it to evaporate and absorb heat from the surroundings.
 5. The evaporator is where the refrigerant absorbs heat from the space or substance being cooled, lowering its temperature.
 6. The refrigerant then flows back to the compressor, where the cycle begins again.
 7. There are various types of refrigerants used in refrigeration systems, including chlorofluorocarbons (CFCs), hydrochlorofluorocarbons (HCFCs), and hydrofluorocarbons (HFCs).
 8. Refrigeration systems can be classified into two main types: vapor compression systems and absorption systems.
 9. Vapor compression systems are the most common type of refrigeration system and use a compressor to compress the refrigerant.
 10. Absorption systems use a heat source to drive the refrigeration cycle and do not require a compressor.

Refrigerating Effect

- Refrigerating effect is the quantity of heat absorbed from a refrigerated space .
- It is the measure of the cooling effect produced by the refrigerant in the evaporator .
- The required refrigerating effect is derived from the equation: $Q_e = m \times C_{p,m} \times \Delta T$, where Q_e is the evaporator load, $C_{p,m}$ is the specific heat of the substance being cooled, and ΔT is the difference between the initial and final temperature of the substance in the evaporator .
- The refrigeration duty, or refrigeration capacity, refers to the total amount of heat absorbed in the chiller by the process, generally expressed in kW .

Ton of Refrigeration

- A ton of refrigeration (TR) is a unit of power used to describe the heat-extraction capacity of refrigeration and air conditioning equipment.
- It is defined as the amount of heat required to melt one ton (2000 pounds) of ice in a 24-hour period.
- One ton of refrigeration is equal to 12,000 British Thermal Units (BTUs) per hour or 3.516 kilowatts (kW).
- The term originated during the early days of refrigeration when ice was the primary means of cooling. The capacity of a refrigeration system was measured by how many tons of ice it could produce in a day.
- Today, the term is still used to describe the cooling capacity of air conditioning and refrigeration systems, although the actual cooling process no longer involves the production of ice.
- In the context of refrigeration and air conditioning, the term “ton” does

not refer to the weight of the equipment, but rather to its cooling capacity.

Coefficient of Performance

The coefficient of performance (COP) is a measure of the efficiency of a refrigeration or air conditioning system. It is defined as the ratio of the heat removed from the refrigerated space to the work input required to achieve this heat transfer.

- The COP is a dimensionless quantity and is typically expressed as a number greater than 1.
- The higher the COP, the more efficient the system is at transferring heat.
- The COP can be used to compare the performance of different refrigeration or air conditioning systems.
- The COP is dependent on the operating conditions of the system, such as the temperature of the refrigerated space and the temperature of the heat sink.
- The COP can be improved by increasing the efficiency of the components of the system, such as the compressor, the condenser, and the evaporator.

In summary, the coefficient of performance is an important parameter in the design and operation of refrigeration and air conditioning systems, as it provides a measure of the efficiency of the system in transferring heat. A higher COP indicates a more efficient system, which can result in lower operating costs and reduced environmental impact.

Unit 3 - Introduction to Refrigeration and Air-Conditioning

Methods of Refrigeration

Refrigeration is the process of removing unwanted heat from a selected object, substance, or space and transferring it to another object, substance, or space. The removal of heat lowers the temperature and may be accomplished by the use of ice, snow, chilled water, or mechanical refrigeration.

Some important methods of refrigeration are: 1. **Ice refrigeration** 2. **Dry ice refrigeration** 3. **Evaporative refrigeration** 4. **Liquid gas refrigeration** 5. **Gas throttling refrigeration** 6. **Air expansion refrigeration** 7. **Vapour compression refrigeration**

The most widely used refrigeration cycle method is **mechanical compression**. It has applications in both air conditioning and commercial and industrial refrigeration. These types of systems transfer heat by mechanically compressing refrigerant into a low-pressure, cold liquid and expanding it into a high-pressure, hot gas.

Cooling caused by the rapid expansion of gases is the primary means of refrigeration today. The technique of evaporative cooling has been known for centuries, but the fundamental methods of mechanical refrigeration were only discovered in the middle of the 19th century.

Unit 3 - Introduction to Refrigeration and Air-Conditioning

Construction and Working of Domestic Refrigerator

A domestic refrigerator works on the principle of vapor compression cycle. The main components of a domestic refrigerator are:

1. **Evaporator:** The evaporator is located in the freezer box in the form of a coil. The liquid refrigerant is evaporated in the evaporator by absorbing heat from the contents of the domestic refrigerator in the cabinet.
2. **Compressor:** The compressor compresses the refrigerant gas.
3. **Condenser:** The coils on the back of the refrigerator let the hot refrigerant gas dissipate its heat.
4. **Capillary tube:** The high-pressure liquid flows through the expansion valve.
5. **Refrigerant:** The system works on closed cyclic operation with the help of a heat transfer fluid called 'refrigerant'.

The working principle of a domestic refrigerator is as follows:

1. The compressor compresses the refrigerant gas.
2. The coils on the back of the refrigerator let the hot refrigerant gas dissipate its heat.
3. The high-pressure liquid flows through the expansion valve.
4. The liquid immediately boils and vaporizes, its temperature dropping to about -25°F .
5. As the cold gas flows through the expansion coils (inside the refrigerator) it makes the inside cold by absorbing heat.

Concept of Heat Pump

A heat pump is a device that transfers heat energy from a source of heat to a destination called a "heat sink". Heat pumps are designed to move thermal energy in the opposite direction of spontaneous heat flow by absorbing heat from a cold space and releasing it to a warmer one. A heat pump uses external power to accomplish the work of transferring energy from the heat source to the heat sink.

Some key points to remember about heat pumps are:

1. Heat pumps work by transferring heat from one place to another.
2. They can be used for both heating and cooling.
3. Heat pumps are more efficient than traditional heating and cooling systems.

4. They can be powered by electricity, gas, or other sources of energy.
5. Heat pumps can be used in a variety of applications, including residential, commercial, and industrial settings.

This is a brief overview of the concept of heat pumps. It is important to understand the basic principles behind heat pumps in order to fully grasp their potential applications and benefits.

Air-Conditioning

Air conditioning is the process of treating air to control its temperature, humidity, cleanliness, and distribution to meet the requirements of the conditioned space. It is an important part of modern life and is used in homes, offices, and public buildings to provide a comfortable and healthy indoor environment.

Some key points to consider when studying air conditioning are:

1. Air conditioning systems can be classified into two main categories: central air conditioning systems and room air conditioners.
2. Central air conditioning systems are used to cool and dehumidify large spaces, such as entire buildings or multiple rooms.
3. Room air conditioners are smaller units that are designed to cool a single room or a small area.
4. Air conditioning systems work by removing heat and moisture from the air, using a refrigeration cycle.
5. The refrigeration cycle involves the circulation of a refrigerant, which absorbs heat from the indoor air and releases it outside.
6. Air conditioning systems also include components such as fans, filters, and ducts to distribute the conditioned air and remove contaminants.
7. Proper maintenance of air conditioning systems is important to ensure their efficient operation and to prevent the growth of harmful microorganisms.

Unit 3 - Introduction to Refrigeration and Air-Conditioning

Meaning and Application

1. **Refrigeration** is the process of removing heat from a space or substance to lower its temperature and maintain it below the ambient temperature.
2. **Air-Conditioning** is the process of treating air to control its temperature, humidity, cleanliness, and distribution to meet the requirements of the conditioned space.
3. The application of refrigeration and air-conditioning is widespread and includes areas such as food preservation, comfort cooling, and industrial processes.
4. Refrigeration and air-conditioning systems use the principles of thermodynamics, heat transfer, and fluid mechanics to achieve their desired effects.

5. The basic components of a refrigeration or air-conditioning system include a compressor, condenser, expansion device, and evaporator.
6. The refrigerant, a substance that absorbs and releases heat, circulates through these components to transfer heat from the space or substance being cooled to the outside environment.
7. The selection of the appropriate refrigerant and system design is critical to the efficient and effective operation of the refrigeration or air-conditioning system.

Unit 3 - Introduction to Refrigeration and Air-Conditioning

Humidity

- Humidity is the amount of water vapor present in the air .
- Water vapor is the gaseous state of water and is generally invisible to the human eye .
- Humidity is usually explained as relative humidity .
- The higher the humidity, the wetter it feels outside .
- Humidity indicates the likelihood for precipitation, dew, or fog to be present .
- Humidity depends on the temperature and pressure of the system of interest .

Dry Bulb Temperature

- Dry Bulb Temperature, usually referred to as air temperature, is the air property that is most commonly used .
- When people refer to the temperature of the air, they are normally referring to its dry bulb temperature .
- The dry-bulb temperature (DBT) is the temperature of air measured by a thermometer freely exposed to the air, but shielded from radiation and moisture .
- DBT is the temperature that is usually thought of as air temperature, and it is the true thermodynamic temperature .
- It is directly proportional to the mean kinetic energy of the air molecules .
- The Dry Bulb Temperature refers basically to the ambient air temperature .
- It is called “Dry Bulb” because the air temperature is indicated by a thermometer not affected by the moisture of the air .
- Dry-bulb temperature - T_{db} , can be measured using a normal thermometer freely exposed to the air but shielded from radiation and moisture .
- The dry-bulb temperature is an indicator of heat content and is shown along the bottom axis of the psychrometric chart or along the left side of the Mollier diagram .

- Constant dry bulb temperatures appear as vertical lines in the psychrometric chart or horizontal lines in the Mollier diagram .

Wet Bulb Temperature

- Wet bulb temperature is the lowest temperature to which air can be cooled by the evaporation of water into the air at a constant pressure.
- It is measured by wrapping a wet wick around the bulb of a thermometer and the measured temperature corresponds to the wet bulb temperature.
- The wet bulb temperature is the temperature of an object that can be achieved through evaporative cooling, assuming good air flow and that the ambient air temperature remains the same.
- By combining the dry bulb and wet bulb temperature in a psychrometric chart or Mollier diagram the state of the humid air can be determined.
- Wet bulb temperature is not the same as Heat Index.
- A comfortable wet-bulb temperature is around 70 °F (22 °C), but no more than 86 °F (30 °C). Above this temperature, physical labor becomes impossible. The higher the wet-bulb temperature, the higher the risk of severe heat stress.

Unit 3 - Introduction to Refrigeration and Air-Conditioning

Dew Point Temperatures

- The dew point temperature, commonly termed dew point or DP, is the temperature to which a parcel of moist air must be cooled at constant atmospheric pressure and constant water vapor content in order for saturation to occur.
- Dew point temperature is defined as, “the temperature of air recorded by a thermometer, when the moisture (water vapor) present in it begins to condense”.
- Dew point depression is “the difference between the dry bulb temperature and dew point temperature of air at any point”.
- Generally, a dew point at or lower than 55 F is comfortable but more than 65 F feels oppressive.
- When you have a high temperature and a high humidity level or dew point, you have a higher heat index as well.
- For example, it may only be 90 F, but it actually feels like 96 because of the high humidity.
- Most people are comfortable with a dew-point temperature of 60 degrees Fahrenheit (16 degrees Celsius) or lower.
- At a higher dew point of, for example, 70 F (21 C), most people feel hot.
- General comfort levels using dew point that can be expected during the summer months: less than or equal to 55: dry and comfortable; between 55 and 65: becoming “sticky” with muggy evenings; greater than or equal to 65: lots of moisture in the air, becoming oppressive.

Comfort Conditions

In Unit 3 - Introduction to Refrigeration and Air-Conditioning of the subject FUNDAMENTALS OF MECHANICAL ENGINEERING, comfort conditions refer to the environmental conditions that are conducive to human comfort and well-being. These conditions include:

1. **Temperature:** The temperature of the air should be maintained within a comfortable range, typically between 20-25°C (68-77°F).
2. **Humidity:** The relative humidity of the air should be maintained within a comfortable range, typically between 40-60%.
3. **Air Movement:** The air should be circulated at a comfortable rate, typically between 0.15-0.25 m/s (30-50 fpm).
4. **Air Quality:** The air should be free of pollutants and contaminants, and should be maintained at a comfortable level of freshness.
5. **Noise:** The noise level should be maintained within a comfortable range, typically below 55 dB(A).
6. **Lighting:** The lighting should be adequate for the tasks being performed, and should be maintained at a comfortable level.

These comfort conditions are important for maintaining the health and well-being of individuals, and are typically achieved through the use of heating, ventilation, and air conditioning (HVAC) systems. These systems are designed to control the temperature, humidity, air movement, air quality, noise, and lighting within a building or space to provide a comfortable and healthy environment for its occupants.

Construction and Working of Window Air Conditioner

A window air conditioner is a small unit with a capacity of 1/2 TOR to 3 TOR, mounted in a window or through a wall . It consists of several components, including a compressor, a condenser, a drier, a filter, a capillary tube, and an evaporator .

The working cycle of a window air conditioner is the vapor compression cycle . An electric motor drives the fan to circulate air through the condenser coil and runs a blower to deliver either cool air to the room or suck air from it before forcing it through the evaporator coil, which cools as well as dehumidifies the air .

The air circuit of a window air conditioner consists of 3 major air circulation systems: room air, outdoor air, and refrigerant air . The air from the room is drawn over the evaporator fan (also called the indoor fan) through a filter, which removes dirt, dust, and impurities. The fan used may be a propeller type or a centrifugal type .

The refrigerant vapor at high pressure and temperature is supplied by the compressor to the condenser. A propeller type fan is used to draw air from the atmosphere and to circulate it through the condenser coil .

Installation of a window air conditioner involves lowering the upper window down on top of the air conditioner, holding it in place, and expanding the air conditioner wings to close the gap on each side . An electrical connection is also required .

Unit 4 - Introduction to Fluid Mechanics and Applications

Fluid mechanics is the branch of physics that studies the behavior of fluids (liquids, gases, and plasmas) and the forces acting on them. It has applications in a wide range of disciplines, including mechanical, civil, chemical, and biomedical engineering, geophysics, oceanography, meteorology, astrophysics, and biology.

Some of the key concepts in fluid mechanics include:

1. **Pressure:** The force exerted by a fluid on a surface per unit area. It is a scalar quantity and is measured in units of pascals (Pa).
2. **Density:** The mass of a fluid per unit volume. It is a scalar quantity and is measured in units of kilograms per cubic meter (kg/m^3).
3. **Viscosity:** A measure of a fluid's resistance to flow. It is a scalar quantity and is measured in units of pascal-seconds ($\text{Pa} \cdot \text{s}$).
4. **Flow rate:** The volume of fluid that flows through a given cross-sectional area per unit time. It is a scalar quantity and is measured in units of cubic meters per second (m^3/s).
5. **Reynolds number:** A dimensionless quantity used to predict the flow behavior of fluids. It is defined as the ratio of inertial forces to viscous forces.

Fluid mechanics has a wide range of applications, including:

- Designing and analyzing fluid systems such as pipelines, pumps, and turbines.
- Predicting the behavior of fluids in natural systems such as rivers, lakes, and oceans.
- Understanding the behavior of fluids in biological systems such as blood flow and respiration.
- Developing new technologies such as microfluidic devices and nanofluidic systems.

In summary, fluid mechanics is a fundamental subject that has numerous practical applications in various fields of science and engineering. It is essential for students to have a solid understanding of the basic concepts and principles of fluid mechanics in order to apply them effectively in their future work.

Introduction to Fluid Mechanics and Applications

Unit 4 of the subject Fundamentals of Mechanical Engineering covers the topic of Fluid Mechanics and its Applications. Fluid mechanics is the branch of physics that studies the mechanics of fluids (liquids, gases, and plasmas) and the forces on them. It has applications in a wide range of disciplines, including

mechanical, civil, chemical, and biomedical engineering, geophysics, oceanography, meteorology, astrophysics, and biology.

Some of the key concepts covered in this unit include: - Properties of fluids: density, viscosity, surface tension, and compressibility - Fluid statics: pressure, buoyancy, and stability of floating bodies - Fluid dynamics: continuity equation, Bernoulli's equation, and flow measurement - Applications of fluid mechanics: hydraulic machines, fluid power systems, and aerodynamics

This unit provides a fundamental understanding of the behavior of fluids and their interactions with solid bodies. It also introduces students to the practical applications of fluid mechanics in engineering and other fields. By the end of this unit, students should be able to analyze and solve problems involving fluid mechanics and apply their knowledge to real-world situations.

Unit 4 - Introduction to Fluid Mechanics and Applications

Fluid Properties

1. **Density:** The mass per unit volume of a fluid is called its density. It is denoted by the symbol ρ . The unit of density is kg/m^3 .
2. **Viscosity:** Viscosity is a measure of a fluid's resistance to flow. It is denoted by the symbol μ . The unit of viscosity is $\text{Pa}\cdot\text{s}$.
3. **Surface Tension:** Surface tension is the property of a liquid that allows it to resist an external force. It is denoted by the symbol σ . The unit of surface tension is N/m .
4. **Vapor Pressure:** Vapor pressure is the pressure exerted by a gas that is in equilibrium with its condensed phases (solid or liquid) at a given temperature in a closed system. The unit of vapor pressure is Pa .
5. **Compressibility:** Compressibility is the ability of a fluid to change its volume in response to a change in pressure. It is denoted by the symbol β . The unit of compressibility is $1/\text{Pa}$.

Unit 4 - Introduction to Fluid Mechanics and Applications

Pressure

- Pressure is defined as the force exerted per unit area.
- The SI unit of pressure is the pascal (Pa), which is equivalent to one newton per square meter (N/m^2).
- Pressure can be measured using various instruments, such as a manometer or a barometer.
- In fluid mechanics, pressure is an important concept as it is used to describe the behavior of fluids at rest or in motion.
- The pressure at a point within a fluid is the same in all directions, this is known as Pascal's principle.

- The pressure in a fluid increases with depth due to the weight of the fluid above it.
- The pressure difference between two points in a fluid can cause the fluid to flow from the point of higher pressure to the point of lower pressure.
- The Bernoulli's equation describes the relationship between the pressure, velocity, and elevation of a fluid in motion.
- Pressure can also be used to calculate the force exerted by a fluid on an object, such as the buoyant force or the drag force.

Unit 4 - Introduction to Fluid Mechanics and Applications

Density

- Density is a measure of mass per unit volume.
- The formula for density is **density = mass/volume**.
- The SI unit for density is kilograms per cubic meter (kg/m^3).
- Density is an important property of fluids, as it affects the behavior of fluids in motion.
- The density of a fluid can change with temperature and pressure.
- An object will float in a fluid if its density is less than the density of the fluid, and will sink if its density is greater than the density of the fluid.
- The density of a fluid can be measured using a hydrometer or by calculating the mass and volume of a sample of the fluid.
- The density of a fluid is used in many calculations in fluid mechanics, including the calculation of buoyancy force and fluid flow rate.

Unit 4 - Introduction to Fluid Mechanics and Applications

FUNDAMENTALS OF MECHANICAL ENGINEERING

Dynamic and Kinematic Viscosity

1. Viscosity is a measure of a fluid's resistance to flow. It describes the internal friction of a moving fluid. A fluid with high viscosity resists motion because its molecular makeup gives it a lot of internal friction. A fluid with low viscosity flows easily because its molecular makeup results in very little friction when it is in motion.
2. Dynamic viscosity, also known as absolute viscosity, is the measure of a fluid's resistance to flow under an applied force. It is commonly expressed in units of Pascal-seconds ($\text{Pa} \cdot \text{s}$) or poise (P).
3. Kinematic viscosity is the ratio of dynamic viscosity to density. It is a measure of a fluid's resistance to flow due to gravity. It is commonly expressed in units of square meters per second (m^2/s) or stokes (St).
4. The viscosity of a fluid can be affected by factors such as temperature, pressure, and the presence of impurities. As temperature increases, the viscosity of a fluid generally decreases. As pressure increases, the viscosity of a fluid generally increases.

5. Viscosity is an important property of fluids in engineering applications. It affects the flow of fluids in pipes, the operation of pumps and other fluid machinery, and the heat transfer characteristics of fluids.

Specific Gravity

Specific gravity is a dimensionless quantity that is defined as the ratio of the density of a substance to the density of a reference substance, typically water at 4°C. It is commonly used to compare the densities of different substances.

Some key points to remember about specific gravity are:

1. Specific gravity is a ratio and therefore has no units.
2. The reference substance for liquids and solids is typically water at 4°C, while for gases it is air at room temperature.
3. Specific gravity is temperature-dependent, as the density of both the substance and the reference substance can change with temperature.
4. Specific gravity can be used to determine the relative density of a substance, which is the ratio of its density to the density of a standard substance, such as water or air.
5. Specific gravity can also be used to determine the buoyancy of an object in a fluid, as an object will float if its specific gravity is less than that of the fluid, and sink if its specific gravity is greater than that of the fluid.

Unit 4 - Introduction to Fluid Mechanics and Applications

Newtonian and Non-Newtonian fluid

1. **Newtonian fluid** is a fluid that has a constant viscosity and flows consistently under applied stress, regardless of the rate of shear. The viscosity of a Newtonian fluid is only affected by changes in temperature or pressure. Examples of Newtonian fluids include water, air, and honey.
2. **Non-Newtonian fluid** is a fluid that does not have a constant viscosity and its flow behavior changes under applied stress. The viscosity of a Non-Newtonian fluid is dependent on the rate of shear, and can increase or decrease with changes in the rate of shear. Examples of Non-Newtonian fluids include ketchup, blood, and quicksand.
3. The difference between Newtonian and Non-Newtonian fluids is important in the study of fluid mechanics and its applications, as it affects the behavior of fluids under different conditions and in different systems.
4. In the subject of FUNDAMENTALS OF MECHANICAL ENGINEERING, understanding the properties and behavior of Newtonian and Non-Newtonian fluids is essential for the design and analysis of fluid systems and their components.

Pascal's Law and Continuity Equation

Pascal's Law

Pascal's Law states that pressure exerted anywhere in a confined incompressible fluid is transmitted equally in all directions throughout the fluid such that the pressure variations (initial differences) remain the same. This principle is stated mathematically as:

$$P = F/A$$

Where: - P is the pressure - F is the force applied - A is the area over which the force is applied

Continuity Equation

The continuity equation is a statement of the conservation of mass. It states that the mass flow rate into a control volume must equal the mass flow rate out of the control volume plus the rate of change of mass within the control volume. Mathematically, this is expressed as:

$$\rho_1 A_1 V_1 = \rho_2 A_2 V_2$$

Where: - ρ is the density of the fluid - A is the cross-sectional area of the flow - V is the flow velocity

The continuity equation is used to analyze fluid flow in pipes and other systems where the fluid is confined.

These concepts are important in the study of fluid mechanics and have many practical applications in engineering. They are covered in Unit 4 - Introduction to Fluid Mechanics and Applications in the subject of FUNDAMENTALS OF MECHANICAL ENGINEERING.

Unit 4 - Introduction to Fluid Mechanics and Applications

Working principles of hydraulic turbines (Pelton Wheel and Francis) & pumps (Centrifugal and Reciprocating) and their classifications and hydraulic lift

- Hydraulic turbines are machines that convert the energy of flowing water into mechanical energy.
- There are two main types of hydraulic turbines: impulse turbines and reaction turbines.
- The Pelton Wheel is an example of an impulse turbine. It operates by converting the kinetic energy of the water into mechanical energy through the use of a series of buckets mounted on the perimeter of a wheel.
- The Francis turbine is a type of reaction turbine. It operates by converting the pressure energy of the water into mechanical energy through the use of a series of fixed and moving blades.

- Hydraulic pumps are machines that convert mechanical energy into hydraulic energy.
- There are two main types of hydraulic pumps: positive displacement pumps and non-positive displacement pumps.
- Centrifugal pumps are a type of non-positive displacement pump. They operate by using a rotating impeller to create a flow of fluid.
- Reciprocating pumps are a type of positive displacement pump. They operate by using a piston or plunger to create a flow of fluid.
- Hydraulic lifts are machines that use hydraulic power to lift heavy loads.
- They operate by using a hydraulic cylinder to apply force to a load, causing it to rise.

Unit 5 - Introduction to Measurement and Mechatronics

1. **Measurement** is the process of determining the value of a physical quantity by comparing it with a known standard.
2. **Mechatronics** is a multidisciplinary field that combines mechanical, electrical, and computer engineering to design and develop advanced systems and products.
3. Mechatronics systems often involve the use of sensors and actuators to measure and control physical quantities.
4. Common measurement techniques include the use of scales, rulers, and gauges to measure length, mass, and pressure, respectively.
5. In mechatronics, measurements are often taken using electronic sensors, which convert physical quantities into electrical signals that can be processed by a computer.
6. Examples of mechatronic systems include robots, automated manufacturing systems, and smart home appliances.
7. The design of mechatronic systems involves the integration of mechanical, electrical, and computer components to achieve a desired functionality.
8. Mechatronics is a rapidly growing field with applications in a wide range of industries, including automotive, aerospace, and consumer electronics.

Introduction to Measurement

Measurement is the process of determining the size, quantity, or degree of something by using a standard unit. It is a fundamental concept in the field of engineering, particularly in mechanical engineering, where precise measurements are crucial for the design, construction, and operation of mechanical systems.

Some key points to consider when discussing measurement in the context of mechanical engineering include:

1. **Units of Measurement:** There are several systems of units used for measurement, including the International System of Units (SI), the United States customary system, and the British imperial system. It is important to be familiar with the units used in a particular context and to be able

to convert between different systems of units.

2. **Measurement Tools:** There are a wide variety of tools used for measurement in mechanical engineering, including rulers, calipers, micrometers, and gauges. The choice of tool depends on the level of precision required and the type of measurement being taken.
3. **Measurement Techniques:** There are several techniques used for taking measurements, including direct measurement, where the object being measured is compared directly to a standard, and indirect measurement, where the measurement is inferred from other measurements.
4. **Measurement Error:** All measurements are subject to some degree of error, and it is important to understand the sources of error and how to minimize them. Sources of error can include the measurement tool, the technique used, and the environment in which the measurement is taken.
5. **Mechatronics:** Mechatronics is the integration of mechanical, electrical, and computer engineering to create advanced systems. Measurement plays a key role in mechatronics, as precise measurements are required for the design and operation of mechatronic systems.

This is a brief introduction to the topic of measurement in the context of mechanical engineering. It is important to have a solid understanding of the principles and techniques of measurement in order to be successful in the field of mechanical engineering.

Concept of Measurement

Measurement is the process of determining the value of a physical quantity by comparing it with a known standard. It is an essential part of science, engineering, and technology.

Here are some key points to consider when discussing the concept of measurement:

1. **Units of measurement:** A unit of measurement is a standard quantity used to express the value of a physical quantity. Common units of measurement include meters for length, kilograms for mass, and seconds for time.
2. **Measurement standards:** Measurement standards are physical representations of units of measurement. They provide a reference against which other measurements can be compared.
3. **Measurement uncertainty:** All measurements have some degree of uncertainty. This uncertainty can arise from various sources, such as the limitations of the measuring instrument or the skill of the person making the measurement.

4. **Accuracy and precision:** Accuracy refers to how close a measurement is to the true value of the quantity being measured. Precision, on the other hand, refers to how close repeated measurements of the same quantity are to each other.
5. **Measurement techniques:** There are many different techniques for making measurements, and the appropriate technique will depend on the quantity being measured and the level of accuracy required.

Error in Measurements

In the context of the Unit 5 - Introduction to Measurement and Mechatronics in the subject of Fundamentals of Mechanical Engineering, error in measurements refers to the difference between the measured value and the true value of the quantity being measured.

There are several sources of error in measurements, including:

1. **Instrumental errors:** These errors arise from the limitations of the measuring instrument, such as the resolution or accuracy of the instrument.
2. **Environmental errors:** These errors are caused by external factors such as temperature, humidity, and air pressure, which can affect the measurement.
3. **Observational errors:** These errors are caused by the observer, such as parallax error or misreading the instrument.
4. **Random errors:** These errors are unpredictable and arise from unknown sources. They can be reduced by taking multiple measurements and averaging the results.

It is important to understand and account for these sources of error in order to improve the accuracy and reliability of measurements.

Calibration

Calibration is the process of comparing the values obtained by a measuring instrument with the corresponding values of a standard. It is an essential step in ensuring the accuracy and reliability of measurements.

Here are some key points to remember about calibration:

1. Calibration is necessary for all measuring instruments, including those used in mechatronics systems.
2. Calibration should be performed regularly to ensure that the instrument is providing accurate readings.
3. The frequency of calibration depends on factors such as the type of instrument, its usage, and the environment in which it is used.
4. Calibration can be performed in-house or by an external calibration service.

5. A calibration certificate is issued after the calibration process, which provides information about the instrument's accuracy and any adjustments made during the calibration process.

Calibration is an important part of the measurement process and is essential for ensuring the accuracy and reliability of measurements in mechatronics systems. It is important to follow the recommended calibration procedures and schedule to ensure that the instrument is providing accurate readings.

Unit 5 - Introduction to Measurement and Mechatronics in the subject of FUNDAMENTALS OF MECHANICAL ENGINEERING

Measurements of Pressure

There are several methods to measure pressure, two of which are the Bourdon Tube Pressure and the U-Tube Manometer.

Bourdon Tube Pressure

- The Bourdon Tube Pressure gauge is a mechanical device used to measure the pressure of liquids or gases.
- It consists of a curved tube, which is flattened and bent into a C-shape or a helix.
- When the pressure inside the tube increases, the tube tends to straighten, causing the movement of a pointer attached to the tube.
- The movement of the pointer is proportional to the change in pressure.

U-Tube Manometer

- The U-Tube Manometer is a simple device used to measure pressure.
- It consists of a U-shaped tube filled with a liquid, usually mercury or water.
- The pressure to be measured is applied to one end of the tube, causing the liquid to rise on that side and fall on the other side.
- The difference in height between the two sides of the tube is proportional to the pressure being measured.

Unit 5 - Introduction to Measurement and Mechatronics in the subject of FUNDAMENTALS OF MECHANICAL ENGINEERING

Temperature (Thermocouple and Optical Pyrometer)

1. A thermocouple is a temperature sensor that consists of two dissimilar metal wires joined at one end, which generates a voltage when the temperature of one of the junctions differs from the reference temperature at the other junction.
2. The voltage produced by a thermocouple is proportional to the temperature difference between the two junctions, and the relationship between voltage and temperature is determined by the Seebeck effect.

3. Thermocouples are widely used in industrial and scientific applications due to their low cost, durability, and wide temperature range.
4. An optical pyrometer is a non-contact temperature measurement device that measures the temperature of an object by detecting the thermal radiation emitted by the object.
5. Optical pyrometers are commonly used in high-temperature applications, such as in the steel and glass industries, where contact temperature measurement is not possible or practical.
6. The accuracy of an optical pyrometer depends on several factors, including the emissivity of the object being measured, the wavelength of the thermal radiation being detected, and the calibration of the instrument.

Mass Flow Rate (Venturi Meter and Orifice Meter)

Mass flow rate is the mass of a substance which passes through a given surface per unit time. It is typically measured in kilograms per second (kg/s). The mass flow rate can be determined using devices such as a Venturi meter or an orifice meter.

Venturi Meter

A Venturi meter is a device used to measure the flow rate of a fluid through a pipe. It consists of a short, converging section followed by a short, diverging section. The fluid's velocity increases as it flows through the converging section, causing a decrease in pressure. The pressure difference between the upstream and downstream sections of the meter can be used to determine the flow rate.

Orifice Meter

An orifice meter is another device used to measure the flow rate of a fluid through a pipe. It consists of a flat orifice plate with a hole in the center, placed in the flow path. As the fluid flows through the orifice, its velocity increases, causing a decrease in pressure. The pressure difference between the upstream and downstream sections of the meter can be used to determine the flow rate.

Strain (Bonded and Unbonded Strain Gauge)

Strain is defined as the deformation of a body due to an applied force. It is a measure of the change in length of a material per unit length. Strain gauges are devices used to measure strain on an object. There are two types of strain gauges: bonded and unbonded.

1. **Bonded Strain Gauge:** A bonded strain gauge is attached directly to the surface of the object being measured. The gauge consists of a wire or foil grid that is bonded to a thin backing material. As the object deforms, the grid deforms with it, causing a change in the electrical resistance of the grid. This change in resistance is measured and used to calculate the strain on the object.

2. **Unbonded Strain Gauge:** An unbonded strain gauge is not attached directly to the object being measured. Instead, it consists of a wire or foil grid that is suspended between two supports. As the object deforms, it causes the supports to move, which in turn causes the grid to deform. This deformation causes a change in the electrical resistance of the grid, which is measured and used to calculate the strain on the object.

Both types of strain gauges are commonly used in the field of mechanical engineering for the measurement of strain in materials and structures. They are an important tool in the study of the mechanical behavior of materials and the design of mechanical systems.

Unit 5 - Introduction to Measurement and Mechatronics in the subject of FUNDAMENTALS OF MECHANICAL ENGINEERING

Force (Proving Ring)

- A proving ring is a device used to measure force.
- It consists of an elastic ring with a known spring constant.
- When a force is applied to the ring, it deforms elastically, and the deformation is measured using a dial gauge or a micrometer.
- The force applied to the ring can be calculated using Hooke's law, which states that the force is proportional to the deformation of the ring.
- Proving rings are commonly used in material testing, such as tensile testing, to measure the force applied to the test specimen.

Torques (Prony Brake Dynamometer)

- A Prony Brake Dynamometer is a device used to measure torque.
- It consists of a brake drum attached to the shaft of the engine or motor being tested, and a brake band that wraps around the drum.
- The brake band is tightened to apply a load to the engine or motor, and the torque is measured by the force required to keep the brake band from slipping.
- The force is typically measured using a spring balance or a load cell.
- Prony Brake Dynamometers are commonly used to measure the torque output of engines and motors.

Concepts of Accuracy

Accuracy refers to the closeness of a measured value to a standard or known value. In the context of measurement and mechatronics, accuracy is an important concept as it determines how well a measurement system can represent the true value of the quantity being measured.

Here are some key points to consider when discussing accuracy in measurement and mechatronics:

1. **Accuracy vs. Precision:** Accuracy and precision are often used interchangeably, but they are not the same thing. Precision refers to the closeness of two or more measurements to each other, while accuracy refers to the closeness of a measured value to a standard or known value.
2. **Sources of Error:** There are many sources of error that can affect the accuracy of a measurement system. These can include systematic errors (such as calibration errors or environmental factors), random errors (such as noise or human error), and gross errors (such as mistakes in recording or reading data).
3. **Calibration:** Calibration is the process of adjusting a measurement system to improve its accuracy. This is typically done by comparing the output of the system to a known standard and making any necessary adjustments.
4. **Error Propagation:** When multiple measurements are combined to calculate a final result, the errors in each measurement can propagate and affect the accuracy of the final result. It is important to understand how errors propagate in order to properly assess the accuracy of a measurement system.
5. **Uncertainty:** Uncertainty is a measure of the range of possible values that a measurement could have, given the sources of error present in the system. It is important to properly quantify the uncertainty in a measurement in order to assess its accuracy.

These are some of the key concepts related to accuracy in the context of measurement and mechatronics. Understanding these concepts is essential for properly designing, calibrating, and using measurement systems in mechanical engineering.

Precision and Resolution

Precision and resolution are two important concepts in the field of measurement and mechatronics, particularly in the subject of Fundamentals of Mechanical Engineering.

- **Precision** refers to the degree of repeatability of a measurement. It is a measure of how close the results of multiple measurements of the same quantity are to each other. A high degree of precision indicates that the measurement process is consistent and produces little random error.
- **Resolution** refers to the smallest change in the quantity being measured that can be detected by the measuring instrument. It is determined by the smallest division or scale interval on the instrument. A high resolution instrument can detect smaller changes in the quantity being measured.

These two concepts are closely related but not the same. It is possible for an instrument to have high resolution but low precision, or vice versa. For example,

a digital scale with a resolution of 0.1 grams may be able to detect small changes in weight, but if it consistently gives different readings for the same object, it has low precision.

In the context of Unit 5 - Introduction to Measurement and Mechatronics, understanding the concepts of precision and resolution is important for selecting and using appropriate measuring instruments and for interpreting measurement data. It is also important for designing and building mechatronic systems that rely on accurate and precise measurements.

Introduction to Mechatronic Systems

Mechatronics is a multidisciplinary field that combines mechanical, electrical, and computer engineering. It focuses on the design and control of advanced systems, such as robots, automation systems, and intelligent machines.

Some key points to consider when studying mechatronics include:

1. Mechatronics is a rapidly growing field that is changing the way we design and control machines.
2. Mechatronics systems are complex and require a deep understanding of multiple engineering disciplines.
3. Mechatronics engineers must be able to work effectively in teams and communicate their ideas clearly.
4. Mechatronics systems often involve the use of sensors, actuators, and control systems to achieve desired performance.
5. Mechatronics is a highly practical field, and hands-on experience is essential for success.

In summary, mechatronics is an exciting and rapidly evolving field that combines multiple engineering disciplines to create advanced systems. It is a challenging but rewarding field that offers many opportunities for innovation and creativity.

Evolution for the notes of the Unit 5 - Introduction to Measurement and Mechatronics in the subject of FUNDAMENTALS OF MECHANICAL ENGINEERING

1. Mechatronics technology helps in lowering development costs, reducing risk, and producing high-quality products .
2. In the design of cars, robots, machine tools, washing machines, cameras, and many other machines, an integrated and interdisciplinary approach to engineering design is increasingly being adopted .
3. Mechatronics brings together areas of technology involving sensors and measurement systems, drive and actuation systems, and microprocessors .
4. The Fundamentals of Metrology seminar introduces participants to the concepts of measurement systems, units, good laboratory practices, data integrity, measurement uncertainty, measurement assurance, traceability,

basic statistics, and how they fit into a laboratory Quality Management System .

5. Transducers and data acquisition systems are covered in Unit 5, including the classification and selection of transducers, resistive, capacitive, and inductive transducers, piezoelectric, Hall effect, optical, and digital transducers, elements of data acquisition systems, smart sensors, and thermal imagers .

Scope for the notes of the Unit 5 - Introduction to Measurement and Mechatronics in the subject of FUNDAMENTALS OF MECHANICAL ENGINEERING

1. Introduction to Measurement and Mechatronics
2. Basic concepts of Measurement and Mechatronics
3. Importance of Measurement and Mechatronics in Mechanical Engineering
4. Types of Measurement and Mechatronics systems
5. Applications of Measurement and Mechatronics in Mechanical Engineering
6. Advantages and disadvantages of Measurement and Mechatronics systems
7. Future scope of Measurement and Mechatronics in Mechanical Engineering
8. Conclusion